

Feature-Enhanced PointPillars for 3-D Millimeter-Wave Object Detection

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PointPillars, a voxel-based 3-D object detection model, would encounter the resolution loss after voxelization, leading to the capability reduction in capturing intricate object details. This limitation is particularly evident in processing 3-D millimeter-wave (MMW) images. Therefore, this article proposes a feature-enhanced PointPillars for 3-D MMW object detection. This enhancement first integrates a multi-scale feature extraction (MFE) module into the pillar feature network. This module is adept at handling the substantial volume of point cloud data characteristic of MMW images and significantly improves feature encoding efficiency. Considering the local density variations and sparsity patterns observed in MMW images, the modified PointPillars further explores a pyramidally attended feature extraction (PAFE) module to improve the inference efficiency. By employing multibranch convolutional kernels with varying dilation rates in the backbone network, the proposed approach expands the receptive fields and augments the contextual interconnectedness of the detected objects. This effectively curtails the semantic and spatial detail loss commonly associated with downsampling. Empirical evaluation of our proposed method against the standard PointPillars benchmark highlights its

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superiority. In particular, our method presents performance enhancement of 0.4% and 7.16% in AP_{R400.5} (AP) for the bird's eye view and 3-D bounding boxes, respectively. Furthermore, it achieves a great 50.4% reduction in the number of parameters and delivers an impressive inference speed of 0.0132 s per frame. These advancements confirm that the augmented network achieves a balance between computational efficiency and 3-D object detection performance for 3-D MMW images, all the while ensuring a practical inference cost.

I. INTRODUCTION

Millimeter-wave (MMW) imaging technique has been widely developed in the field of security inspection [1], [2], [3], [4]. To rapidly and accurately detect the concealed objects is one of the key problems. However, 3-D MMW images are usually noisy and have low resolution, while mostly the concealed objects are small in size and easy to be confused by human background. These issues, combined with the big-data feature of 3-D MMW images, make the performance of object detection challenging [5].

Recently, 3-D object detectors based on LiDAR point cloud have made great breakthroughs [6], [7], [8], [9]. Shi et al. [10] proposed Point Voxel-RCNN (PV-RCNN), a novel method for accurate 3-D object detection from point clouds. This method deeply integrates both 3-D voxel convolutional neural network (CNN) and PointNet-based set abstraction to learn more discriminative point cloud features. Moreover, it takes advantage of efficient learning and high-quality proposals of the 3-D voxel CNN and the flexible receptive fields of the PointNet-based networks, significantly improving the 3-D object detection performance compared with previous state-of-the-art methods. Deng et al. [11] found that precise positioning of raw points is not essential for high performance of 3-D object detection and that the coarse voxel granularity can also offer sufficient detection accuracy. On this basis, a detection framework named Voxel R-CNN was proposed, which strikes a careful balance between accuracy and efficiency by taking full advantage of voxel representations to characterize point clouds. Liu et al. [12] proposed a MA-MFFC detector that explores the multiattention mechanism (MA) and the multiscale feature fusion network with ConvNeXt module (MFFC) to comprehensively improve the detection performance of the network. Specifically, the MA module contains point-channel attention and voxel attention. By taking pointwise and channelwise features into consideration, the MA enhances the information representation of key points in voxels, suppresses background point clouds in voxelization, and improves the robustness of the network. The MFFC module can extract rich feature information, in which the conventional convolution layer is replaced by the ConvNeXt module to enhance the feature extraction capability of the network.

On the other hand, Lang et al. [13] proposed PointPillars, a novel encoder that utilizes PointNets to learn a representation of point clouds organized in pillars. The encoder learns features on pillars of the point cloud to predict 3-D oriented boxes for objects, which can greatly leverage the full information represented by the point cloud.

Furthermore, a lean downstream network in PointPillars was proposed to enable a tradeoff between speed and accuracy, outperforming previous encoders with respect to both speed and accuracy by a large margin. These improvements show PointPillars is an appropriate architecture for 3-D object detection with high detection performance. Therefore, PointPillars has increasingly become one of the most widely used models in 3-D object detection. Tao et al. [14] proposed PointPillar+CBAM target detection algorithm, which achieves the amplification of local information in the three scale feature maps and better extracts the more important feature information. Compared with the benchmark, this algorithm enhances the PointPillar target detection performance and improves the accuracy of target prediction. Xu et al. [15] proposed an improved laser point cloud object detection method. Three key improvements were introduced based on PointPillars: Softplus activation function, attention mechanism SOPA (pillar second order of point attention) and increased positive sample weight, which effectively improve the detection ability of important objects. Then, Zhong et al. [16] proposed an efficient sampling weighted fusion encoder for 3-D object detection. This encoder introduces an efficient point cloud sampling technique that selects relevant points for each pillar, reducing computational complexity without compromising detection accuracy. Additionally, a pillar feature learning network named WFNet is developed to extract essential point information. Moreover, the efficient sampling weighted fusion encoder (SWFNet) is incorporated into the PointPillars framework as an encoder replacement, which significantly improves the detection performance of small objects.

Therefore, existing inference models have demonstrated good detection performance in the field of Lidar, exhibiting their great potential for 3-D MMW object detection. Drawing inspiration from PointPillars, this article aims to assess its potential for effective use in the detection of objects within 3-D MMW imagery. The PointPillars architecture facilitates computational efficiency during convolutional operations, which is a benefit derived from structured data storage and the voxelization process. However, the voxelization step is not without its tradeoffs, particularly in terms of information loss, which is contingent on voxel resolution selection. To counteract this and to concurrently achieve high-efficiency and high-accuracy inference, this article proposes a feature-enhanced PointPillars, especially within MMW conditions. First, a multiscale feature extraction (MFE) module is integrated into the feature encoding stage of the pillar feature network. This enhancement amplifies the feature aggregation capacity within the pillars of the point cloud data and the current point pillars thereby can retain the 3-D features of MMW images to a great extent. Besides, the model's computational speed is significantly augmented during the conversion from 3-D point clouds to pseudoimages. To further refine the feature extraction capability of the backbone network in PointPillars, a pyramidally attended feature extraction (PAFE) [17] is adopted and integrated in the backbone network. Combined with the multiscale fusion strategy, this module

employs multibranch convolution kernels with varying dilation rates to introduce different receptive fields. This alleviates the degradation of detection accuracy caused by semantic and spatial information loss due to internal down-sampling of the backbone network. It substantially reduces the loss of detail information of objects of small size caused by the voxel resolution and the encoding method.

This dual integration of MFE and PAFE advances PointPillars, rendering it a more robust solution to extract and utilize the details in MMW images. Also, it can achieve a balance between computational efficiency and detection performance. We evaluated our enhanced PointPillars network on the KITTI test method and ablation experiments, with evaluation indexes including bird's eye view (BEV) perspective, 3-D box, and orientation similarity. Experimental results show that our proposed method maintains a high score with the BEV AP_{R40,5} and AOS AP_{R40,5} at 0.964 and 0.979, respectively. The 3-D AP_{R40,5} has increased by 7.16%, effectively improving the locating detection performance of small targets.

II. FEATURE-ENHANCED POINTPILLARS

The diagram of the feature-enhanced PointPillars is shown in Fig. 1, where the basic three stages of the benchmark PointPillars, i.e., pillar feature net, backbone, and detection head, are retained. The proposed method focuses on exploring the multiscale feature learning technique in the pillar-feature-net stage and the backbone stage to enhance the learning ability of features in pillars and in pseudoimages, respectively. Besides, the detection head is consistent with the standard PointPillars.

A. Multiscale Feature Extraction Integrated in Pillar Feature Net

Since PointPillars takes less consideration about the specific position information of each point, it may result in the loss of part of the spatial geometric distribution information and make it difficult for subsequent object detection. Therefore, this article first proposes to integrate the MFE module into the pillar feature net, and its detailed flowchart is shown in Fig. 2.

After pillaring, the feature of columnized point cloud of size (D, P, N) is obtained, where D , P and N denotes the dimension of point feature vectors, the number of nonempty pillars per sample and the number of points per pillar, respectively. Then a transformation network implemented by PointNet is exploited to extract and transform this feature, yielding a point cloud pillar, denoted as $\mathbf{P}_0$ with a dimension of (C, P, N) , where C denotes the channel number. The i th pillar of $\mathbf{P}_0$ is then sliced out and denoted as $\mathbf{P}_i$ with the dimension of $(C, 1, N)$, where the subscript i refers to the i th pillar, $i = 1, \dots, P$. The max pooling is performed on $\mathbf{P}_i$ to get the global feature vector $\mathbf{p}_i$, which has a dimension of $(C, 1, 1)$. Subsequently, taking $\mathbf{p}_i$ as the center, the Euclidean distance from any point feature vector $\mathbf{p}_{ij}$ in the pillar $\mathbf{P}_i$ to the global feature $\mathbf{p}_i$ is computed traversically to obtain the distance vector $\mathbf{d}_i$, where $\mathbf{p}_{ij}$ has the dimension of

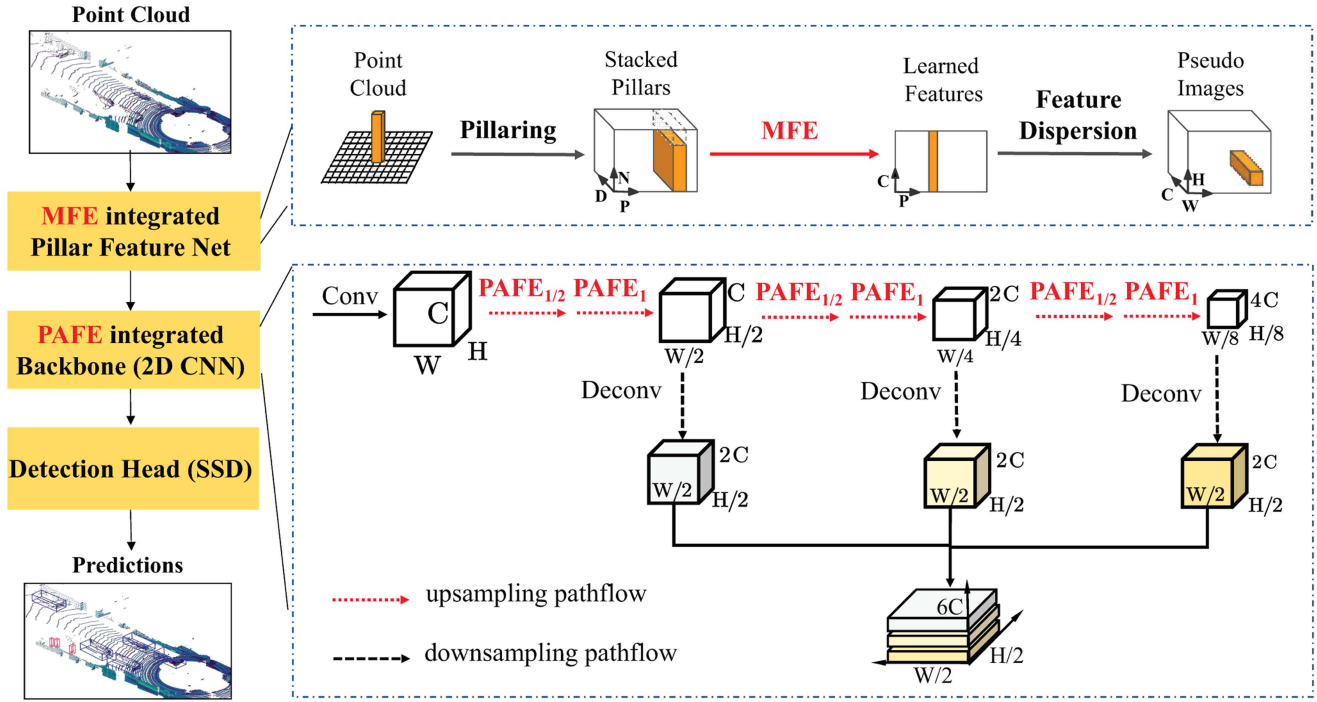


Fig. 1. Feature-enhanced PointPillars. MFE: Multiscale feature extraction, PAFE: Pyramidally attention feature extraction.

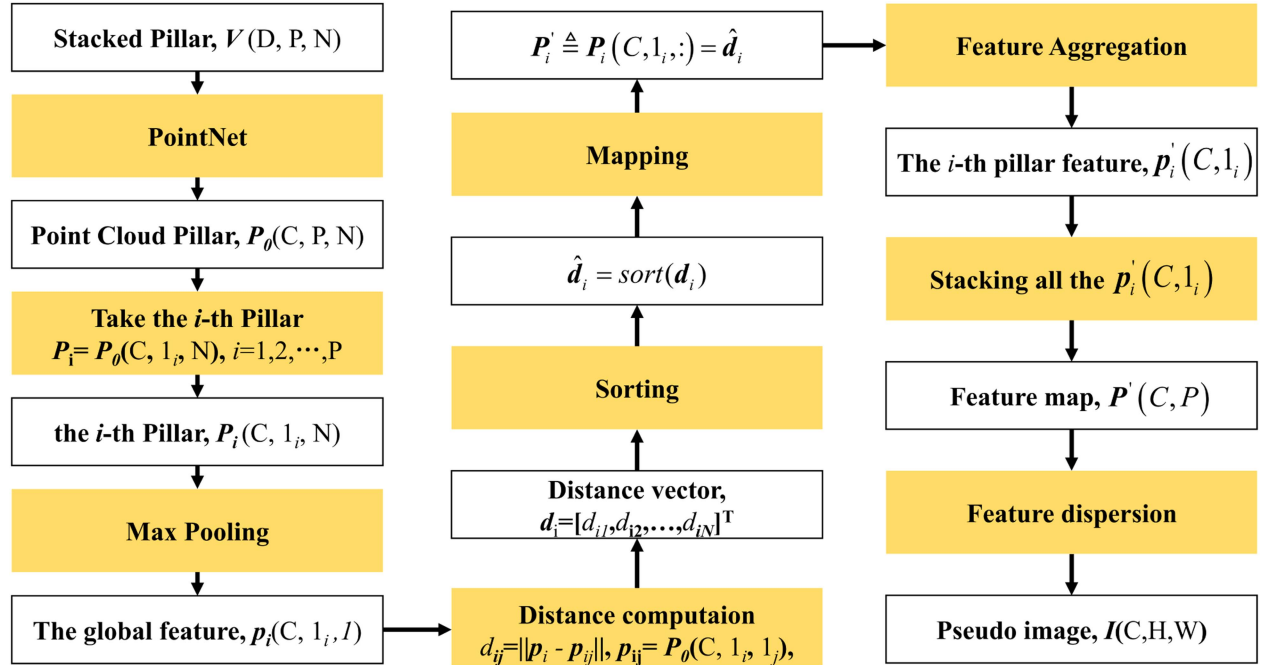


Fig. 2. Flowchart of the integrated MFE module.

$(C, 1_i, 1_j)$, j denotes the j th vector of P_i and $j = 1, \dots, N$. After sorting d_i , the result is remapped back to P_i , to update the feature pillar P'_i with the dimension of $(C, 1_i, N)$, which is represented as the feature of the point cloud pillar in the corresponding scale. By searching the nearest neighbor points with different number (4, 8, 16, 32), the features of various scales are spliced and aggregated, which then are fed

into the fully connected layer network for feature learning. Then, the feature representation in the current pillar, with a dimension of $(C, 1_i)$, that is p'_i , is output. All the p'_i are then stacked to export the tensor of (C, P) and these features are finally dispersed to the location of the original point pillar, so as to obtain the pseudoimage of size (C, H, W) , where H and W denote the height and width of the canvas, respectively.

TABLE I
Detail Parameters of PAFE Module

Module	Convolution type	Input-output channel	Convolution kernel	Step length	Padding	Packet number	Voidage	Feature map
branch1	depthwise separable	(64,128)	3	2	1	2	1	(128, 248,216)
branch2	depthwise separable	(64,128)	(3,3)	2	1	2	2	(128, 248,216)
branch3	CBAM [18]	(128,128)	–	–	–	1	1	(128, 248,216)
	depthwise separable	(64,128)	(3,3)	2	1	2	4	(128, 248,216)
	CBAM	(128,128)	–	–	–	1	1	(128, 248,216)
branch4	global average pooling	(64,64)	(248,216)	–	0	2	1	(128, 248,216)
	upsampling	(64,128)	–	–	0	1	–	(128, 248,216)

Based on the above feature coding mode, the feature vector of each pillar can be obtained, and the information involved in pillar can be enhanced, and the loss of local geometry information of the 3-D point cloud during the traditional feature learning phase can be alleviated.

B. Pyramidally Attended Feature Extraction Integrated in Backbone Network

As shown in Fig. 1, the pseudoimage generated by the pillar feature net is channeled into a 2-D convolutional network that constitutes the backbone of the model. This backbone network bifurcates into two distinct pathways: downsampling and upsampling. Deviating from the traditional feature extraction methodologies employed by the standard PointPillars model, this framework harnesses a PAFE module within the down-sampling pathway. The PAFE module supersedes standard convolutions, augmenting the extraction of features by attending to multiple receptive fields across varied granular levels. As depicted in Fig. 1, the PAFE module operates in sequential pairs: the first module, setting the standard convolution stride to 2, reduces the feature map dimensionality by half, while the second retains the feature map size by adjusting the stride to 1. This dual-stage PAFE deployment enables a more refined and hierarchical feature representation, facilitating a more robust feature extraction mechanism that is sensitive to both global and local spatial contexts within the input data.

The detailed diagram of PAFE is shown in Fig. 3. This module makes use of a combination of convolutions and the attention mechanism to enrich the feature maps with context information at different scales, allowing the model to gain a broader view and capture nearly global contextual details. Specifically, the module incorporates different receptive fields within each feature flow path, allowing it to learn and integrate features at various scales. This is achieved by using convolutional layers with different kernel sizes (e.g., 1×1 and 3×3) and dilations ($D = 2$, $D = 4$), which helps in capturing both fine and coarse details within the input. Furthermore, by utilizing a broader field of view, the module can gather more contextual information. This

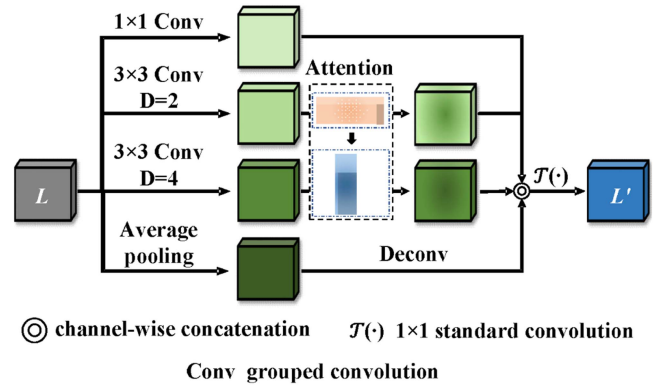


Fig. 3. Schematic diagram of the PAFE module.

is beneficial for understanding the scene as a whole and assists in accurately locating and classifying objects within the scene. The inclusion of an attention component within the module focuses the learning on relevant features by weighting the importance of different areas in the feature map. This targeted approach helps in distinguishing between the relevant and irrelevant parts of the data, which is crucial for precise object detection. The nearly global perspective provided by the PAFE module aids the detection head in making localized decisions with a comprehensive understanding of the entire scene, thereby increasing the overall accuracy of the whole model. The parameters in the depth separation convolution layer of each Conv from top to bottom are shown in Table I.

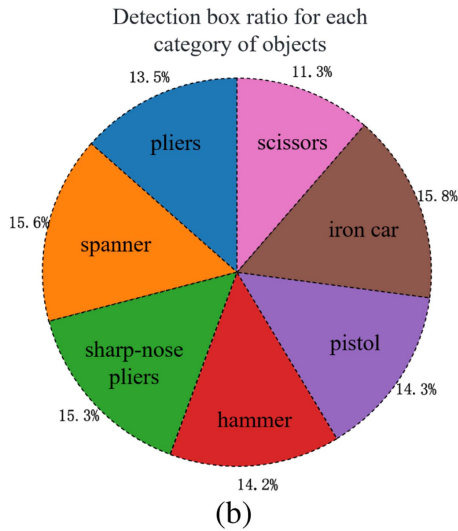
III. EXPERIMENTS

A. Dataset

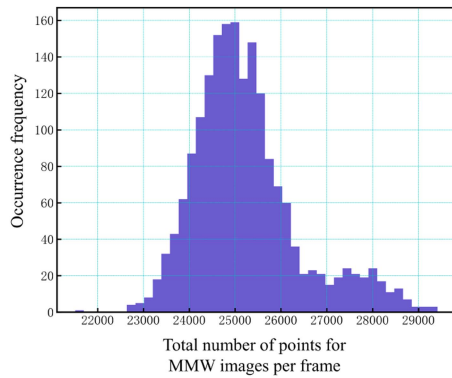
In the field of 3-D MMW object detection, the lack of comprehensive public data sets is a significant obstacle. Therefore, a homemade dataset of human concealed objects by the laboratory-developed 3-D MMW imaging system is prepared. The dataset encompasses a spectrum of items—pliers, a spanner, sharp-nose pliers, a hammer, a pistol, a die-cast car, and scissors—as depicted in the Fig. 4(a). The MMW imaging system operates at a carrier frequency of



(a)



(b)

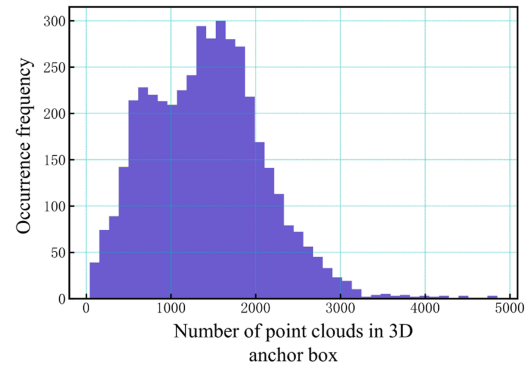


(c)

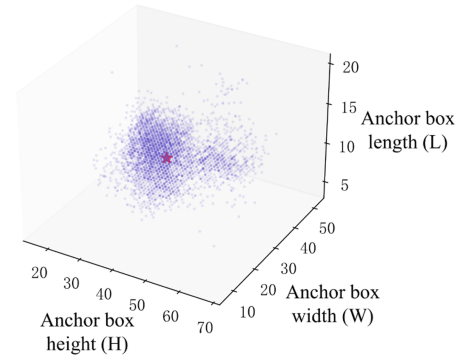
Fig. 4. (a) Objects for actual detection. (b) MMW point cloud image object category proportion. (c) Point cloud data distribution.

35 GHz with a bandwidth of 4 GHz. The data collection process involved scanning five individuals of varied heights and body types to construct the dataset.

The obtained 3-D MMW point cloud image is first projected along the depth direction to yield a frontal view image. The 2-D bounding boxes and class labels for the concealed objects within the projected image are annotated based on the principles of the pinhole camera model, facilitating the establishment of pertinent parameters. Building upon these annotations, the derivation of 3-D bounding boxes is undertaken. Specifically, during the translation from 2-D annotations to 3-D space, a uniform scaling factor of 1:1 is applied to the width and height dimensions to simplify computation. For the depth dimension, a methodical approach is employed where the mean intensity value across each layer of the point cloud is calculated. These mean values are then normalized, and a threshold ratio of



(a)



(b)

Fig. 5. Distribution statistics of MMW point cloud images and 3-D bounding box data in the form as (maximum, minimum, mean, variance). (a) Point statistics (4851, 41, 1414, 466518). (b) Length, width, and height statistics: Height (68, 16, 41.25, 60.68), width (52, 7, 24.44, 32.38), length(20, 4, 12.47, 3.57).

0.05 is set to exclude layers exhibiting minimal intensity, thereby filtering out noise and irrelevant data. This process enables the precise determination of the depth indices at the proximal and distal ends of the concealed objects, which in turn promotes the accurate estimation of the objects' depth range and their spatial coordinates within the 3-D bounding box framework.

Therefore, the dataset includes a total of 4366 labeled 2-D/3-D truth bounding boxes that encapsulate hidden object instances. Among them, the percentage distribution of detection box for each category is shown in Fig. 4(b). On the whole, the sample boxes of object categories are in a balanced distribution and the proportion of values ranges from 11.3% to 15.8%. As shown in Fig. 4(c), the total number of points per frame point cloud image in the dataset can be extended from 21 509 to 29 418, and the median point value is 25 032, without the occurrence of extreme values. Also, it reflects that in the same detection scene, the imaging foreground extraction method based on clustering is effective. The detailed data distribution of sample boxes is shown in Fig. 5, and all coordinate scales are measured in pixel intervals. In Fig. 5(a), the distribution of point counts within real anchor boxes ranges from 41 to 4851, with the majority of boxes containing point counts ranging from 730 to 2097. According to the classification of easy, moderate,

and hard samples in KITTI format [19], [20] and without considering truncation and other additional conditions, the corresponding sample box values of our dataset are 2103, 2225, and 0, respectively. In addition, there are 38 sample boxes with image pixel height less than 25, yielding a total of 4366 sample boxes. In the subsequent training process of the detection model, this article controls the size of the training set and the test set with a ratio of 7:3.

B. Evaluation Index

In this section, the precision (P) and recall (R) are taken as metrics to evaluate performance in MMW image detection tasks. The precision-recall (PR) curve is employed to derive the Average Precision (AP), which serves as an indicator of accuracy. For a given intersection over union (IoU) threshold, we apply three distinct evaluation indices to gauge the degree of congruence between the model's 3-D predicted bounding boxes and the ground truth.

The first index, BEV AP_{R40,0.5}, quantifies the accuracy of overlap when the 3-D detection boxes and the ground truth are projected onto a 2-D plane from a BEV. This measure reflects the precision of detection in the plane parallel to the ground. The second index, 3-D AP_{R40,0.5}, assesses the accuracy of the predicted bounding box in 3-D space in relation to the actual 3-D ground truth. The third index, AOS AP_{R40,0.5}, measures the average orientation similarity (AOS) [21], which is an evaluation of how closely the predicted average orientation angle of the detected object matches the ground truth across the dataset. These indices provide a comprehensive evaluation framework for the accuracy of detection and orientation estimation in 3-D MMW imaging systems.

In the process of computing AOS, we adopt the updated 40-point AP_{R40} methodology. This approach is an enhancement of the conventional 11-point interpolation used for calculating AP metrics [22]. The specific formula is shown in the following:

$$AP|_{R40} = \frac{1}{N} \sum_{r \in \{1/40, 2/40, \dots, 1\}} \max_{\tilde{r}: \tilde{r} \geq r} s(\tilde{r})$$

$$s(r) = \frac{1}{|\mathcal{D}(r)|} \sum_{i \in \mathcal{D}(r)} \frac{1 + \cos \Delta_{\theta}^{(i)}}{2} \delta_i \quad (1)$$

where the orientation similarity $s \in [0, 1]$ at recall r is a normalized ($[0..1]$) variant of the cosine similarity, $\mathcal{D}(r)$ represents the set of all predicted positive samples under the recall rate r , and $\Delta_{\theta}^{(i)}$ represents the difference between the predicted angle of detected object i and the truth marked box. In order to punish multiple checks for matching the same truth marked box, $\delta_i = 1$ would be set if the check i matches the truth marked box, otherwise $\delta_i = 0$. This finer sampling of recall levels enables a more detailed and accurate representation of the precision-recall relationship, providing a robust metric for the evaluation of detection algorithms in MMW imaging.

In addition to AP, the frames per second (FPS), denoting the inference time for a single frame across different algorithm models on the same device, is adopted to evaluate the detection speed, as shown in the following:

$$FPS = \frac{1000}{\text{time}}. \quad (2)$$

C. Parameter Setting

In this article, the experimental hardware configuration is RTX 4090, and the adopted benchmark model and subsequent optimized model are both based on OpenPCDet [23] framework, and the network input is the homemade dataset of 2000 3-D point cloud images.

To update the network gradients, the Adam-M optimizer along with a one-cycle learning rate policy is employed for network training from the ground up. This training utilizes a peak learning rate of 0.0003, a division factor of 10 for annealing the learning rate, a weight decay of 0.01 to prevent overfitting, and a momentum of 0.9 to stabilize convergence. The batch size is set at 8. Throughout the training phase, approximately 12.7 GB of video memory is consumed, and the process spans 80 epochs. The data augmentation strategy implemented is consistent across all models and follows the PointPillars framework. Initially, for each training set frame, a lookup table is constructed that delineates the correlation between each annotated 3-D ground truth bounding box and the corresponding points within the point cloud. Subsequently, 20 ground truth boxes are randomly selected and integrated into the current point cloud for each training sample. A collision test is conducted to eliminate any overlapping bounding boxes that would result in physically implausible configurations. Each marked ground truth bounding box, along with its associated point cloud data, is subjected to probabilistic data augmentation. Specifically, postintegration of each new box, the training sample is randomly rotated and translated to diversify the training dataset. The rotation is uniformly sampled within the range of $[-\pi/2, \pi/2]$ about the Z-axis. Additionally, random flipping along the X-axis is applied with a 50% probability, and the data are randomly scaled with magnification factors selected from the interval $[0.95, 1.05]$, thereby enhancing the robustness of the model to variations in object orientation and scale.

The sizing of anchor boxes in the model is based on the statistical distribution of the 3-D bounding boxes, as illustrated in Fig. 5(b). The mean pixel dimensions for length, width, and height are translated into the corresponding physical dimensions in the model's coordinate system, with considerations for each network's specific scaling factors. For both the PointPillars network and its derivative architectures, a standardized anchor box dimension of (0.4, 0.24, 0.13) meters is assigned along the X-, Y-, and Z-axes, respectively. Moreover, in alignment with the input resolution and the scale of the modeled environment, the voxel grid is partitioned with dimensions of (0.04, 0.04, 0.7) m. Utilizing the statistical mean values for depth and length

TABLE II
3-D Detection Results of MMW Images Based on KITTI Test Method

Detection algorithm \ Evaluation index	BEV AP _{R40_{0.5}}	3D AP _{R40_{0.5}}	AOS AP _{R40_{0.5}}	Parameter number(M)	FPS ⁻¹	Floating-point number(GFLOS)
VoxelNet[24]	0.9382	0.8542	0.9782	6.42	0.2308 s	312.97
SECOND[19]	0.8954	0.7469	0.9804	4.59	0.012 s	76.44
PointPillars[13]	0.9642	0.8655	0.9821	4.82	0.0076 s	62.25
Proposed	0.9598	0.9277	0.97920	2.39	0.0132 s	70.55

from the dataset, the base of the 3-D anchor box is positioned at a height of 0.06 m from the reference plane.

During the matching phase of anchor box assignment, to diminish the rate of false positives while ensuring precise regression, the thresholds for matching an anchor box to a positive sample (an actual object) or a negative sample (background or nonobject space) are judiciously set. A threshold of 0.6 is used for positive sample matching, recognizing the geometric and spatial similarity between the detected objects and certain parts of the human anatomy. Conversely, a lower threshold of 0.45 is applied for negative sample matching. This dual-threshold strategy serves to optimize the balance between sensitivity and specificity, thereby enhancing the overall accuracy and reliability of the detection model.

D. Experimental Results

A horizontal comparison of MMW image detection performance is first conducted among the VoxelNet [24], SECOND [19], PointPillars [13], and our proposed method. The evaluation indexes in it are inference speed (i.e., FPS), BEV [25], 3-D box intersection ratio, directional coincidence degree, parameter number, and floating point computation. The score threshold is set to 0.1, while the the IoU value is 0.5.

From the first three evaluation indices in Table II, it shows that the proposed algorithm has achieved optimal evaluation results in BEV (i.e., BEV AP_{R40_{0.5}}), 3-D anchor box (i.e., 3-D AP_{R40_{0.5}}), and angle orientation (i.e., AOS AP_{R40_{0.5}}) of MMW point cloud images. From the BEV prospective, the AP of the proposed algorithm has an accuracy improvement of 7.19% compared with the lowest SECOND algorithm, which is only slightly different from its base model PointPillars by 0.0053. Combined with the calculation process of BEV and the method for preparing 3-D anchor boxes of our dataset, it clarifies that the upper limit of BEV-AP for binary classification model learning is close to 0.96 in the MMW image detection scene. It implies that it is less effective to improve the accuracy in BEV by optimizing the model. Besides, compared with the slight improvement of the accuracy in BEV, the AP on the 3-D anchor boxes of the proposed algorithm, i.e., 3-D AP_{R40_{0.5}}, has an increase of 8.6%, 24.21%, and 7.19% compared with the other three, respectively. This

result demonstrates that our improved network has a more accurate location in the depth direction, which performs the superiority of pillar coding, compared with the simple voxelization of point cloud for object detection and location. For the inference of anchor box direction, due to the uniform setting of 0° orientation in the dataset, the network learning in the anchor box direction is extremely simple, and the AOS AP_{R40_{0.5}} scores of different networks are greatly high.

From the latter three evaluation indices of Table II, it is obvious to have that although there is little difference in the number of different network parameters (referring to the parameter-number column), there is a great difference in floating point arithmetic (referring to the FPS⁻¹ column and the floating-point-number column). VoxelNet, the first model to apply the idea of voxel to 3-D detection, generates a large number of empty voxel representations in the process of scene partitioning, resulting in a huge amount of computing consumption in the feature extraction stage and a significant reduction in its detection speed. Such that VoxelNet has to consume 0.2308 s per frame, i.e., FPS⁻¹ = 0.2308s. SECOND, an optimized VoxelNet model that uses sparse convolution instead of 3-D standard convolution, reduces the floating point number by 3.1 times and increases the inference speed by 18.2 times compared to VoxelNet. Both the proposed algorithm and its base model PointPillars adopt converting point cloud into 2-D pseudoimages after pillaring. The proposed algorithm, however, utilizes the multibranch superposition in the feature extraction stage. This will increase the floating-point number slightly to improve inference performance. Therefore, compared to its base model, the detection speed of the proposed algorithm is increased by 0.0056 s while the parameter number is decreased by 50.4%.

Fig. 6(a)–6(c) shows the transformation curves of AP about IoU threshold from the BEV perspective of the network, 3-D anchor box and AOS, respectively. Fig. 6(d) is an auxiliary illustration of the emergence of the equivalent distribution in Fig. 6(c). The higher accuracy shows that the location performance of the network is better. According to Fig. 6(a) and 6(b), each model at first maintains a high accuracy at the small threshold phase, and then with the increase of the IoU threshold, the accuracy begins to decline until it reaches 0. However, compared with the other

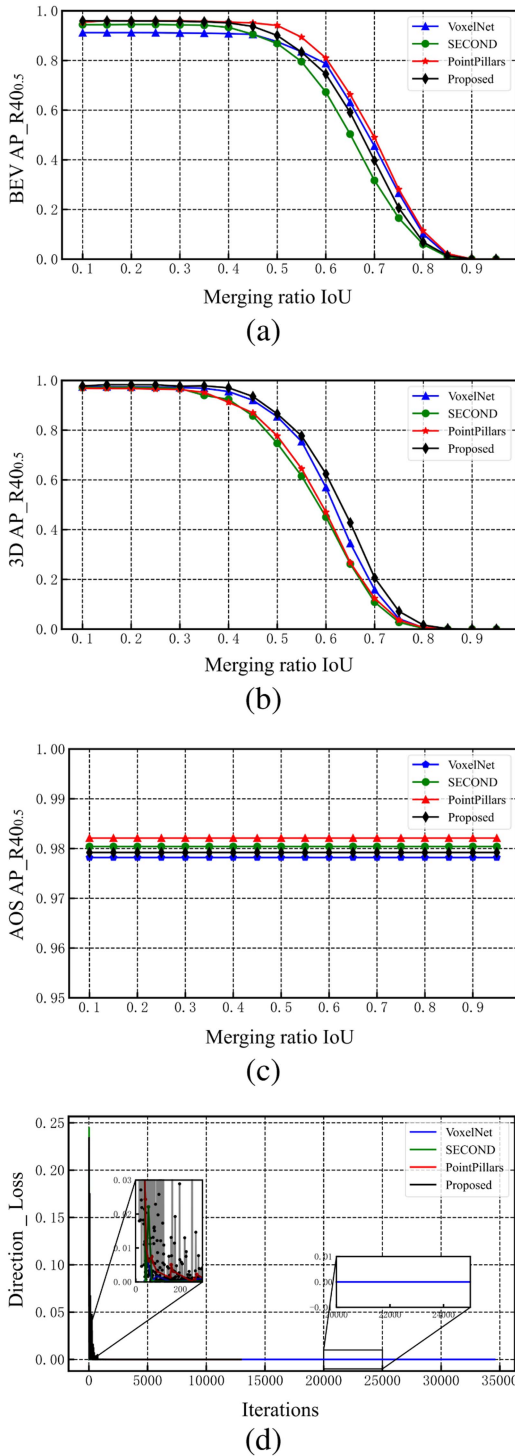


Fig. 6. AP versus IoU curve for different models. (a) For BEV perspective. (b) 3-D box calculation. (c) AOS orientation calculation. (d) Convergence status of direction loss for different models during training.

three algorithms, the proposed algorithm always gets the highest score in the process of index decline, which also indicates that the enhanced algorithm effectively improves the locating accuracy of the network. Within the range of the IoU threshold, Fig. 6(c) shows that the scores of AOS AP_R40 of all networks remain very similar. To further

confirm the performance of AOS, the convergence process of direction loss during training is examined, as shown in Fig. 6(d). It can be seen that at the beginning of the iteration, the curves of different algorithms do change when fitting and learning 3-D anchor box orientation. However, as the number of the iteration increases, all networks converge quickly and complete the learning of the spatial features of anchor box orientation.

In order to verify the importance of the MFE module and the PAFE module, the ablation experiments are conducted before and after the improvement of the benchmark model, and the results are shown in Table III. Although the introduction of the MFE module has increased the floating point computation amount by 8.3G B, the PAFE module has been adopted in backbone, reducing the number of module cycles and the parameters of the model by about 50.4%. The model inference speed is slightly increased by 0.0097 s. The results of three kinds of AP correspond well to those in Fig. 7. The ablation experiments strongly prove that the redesigned MFE module in feature coding and the MFE module PAFE in backbone can indeed improve the detection performance of the proposed model greatly.

In order to obtain a suitable score threshold for visualization of the ablation experiments, the score threshold is taken as the independent variable and the distribution of the inference detection boxes of the proposed algorithm is taken as the dependent variable to draw detection images, as shown in Fig. 7. With the gradual increase of the score threshold, the redundant prediction boxes are completely eliminated in the point cloud image of this frame until it reaches about 0.4, so 0.4 is set as the baseline threshold.

Therefore, Fig. 8 gives a visual representation of the detection under the conditions that the score and IoU thresholds are 0.4 and 0.5, respectively. As shown in the second column of Fig. 8, the object detection result of the benchmark PointPillars is not ideal. On the one hand, it is difficult to distinguish the shape of part of the human torso and the objects in point cloud after imaging by MMW imaging system. On the other hand, spatial coding has more dimensions than the reflection intensity coding in point cloud detection, which makes feature extraction more dominant. So the combination of the two factors results in redundant reasoning boxes and trunk false detection. The comparison of ablation experiments is shown in Fig. 8 from the second column to the fifth column. Among them, the third column shows that after adopting the MFE module, the feature calculation of redundant dimensions is reduced through backpropagation learning, and the appearance of redundant boxes is reduced to a certain extent. With the integration of the PAFE module, the fourth and fifth columns in Fig. 8 show that the multibranch voiding-rate field feature multiplexing module acquires more context information around the object, further assists in judging the object and the location between the objects, and optimizes the detection accuracy of PointPillars. The accuracy and effectiveness of the proposed algorithm in object detection are verified.

TABLE III
Results of Ablation Experiment

Detection algorithm Evaluation index	PointPillars	PointPillars	PointPillars	Proposed
Point feature net (MFE)	—	✓	—	✓
Backbone (PAFE)	—	—	✓	✓
Floating-point number (GFLOPs)	62.25	62.23	70.57	70.55
Parameter number (M)	4.82	4.81	2.39	2.39
FPS ⁻¹	0.0067 s	0.0072 s	0.0128 s	0.0132 s
BEV AP _{R400.5}	0.960	0.956	0.958	0.964
3D AP _{R400.5}	0.866	0.887	0.912	0.928
AOS AP _{R400.5}	0.982	0.972	0.976	0.979

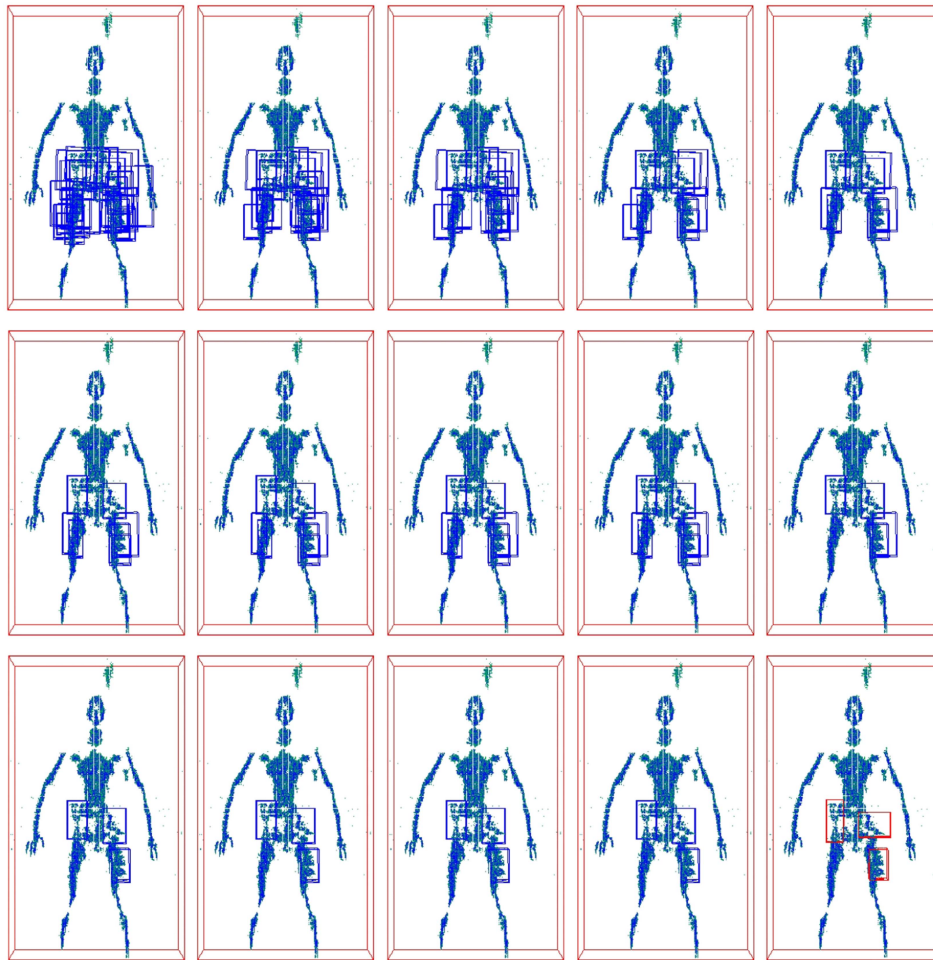


Fig. 7. Detection results of different scoring thresholds. From left to right, the images in the first row represent detection results with score thresholds of 0.01, 0.02, 0.03, 0.04, and 0.05, respectively. The images in the second row from left to right are the detection results with thresholds of 0.06, 0.07, 0.08, 0.09, and 0.1, respectively. The third row shows the test results with score thresholds of 0.2, 0.3, 0.4, and 0.5, respectively, and the last image denotes the ground-truth box.

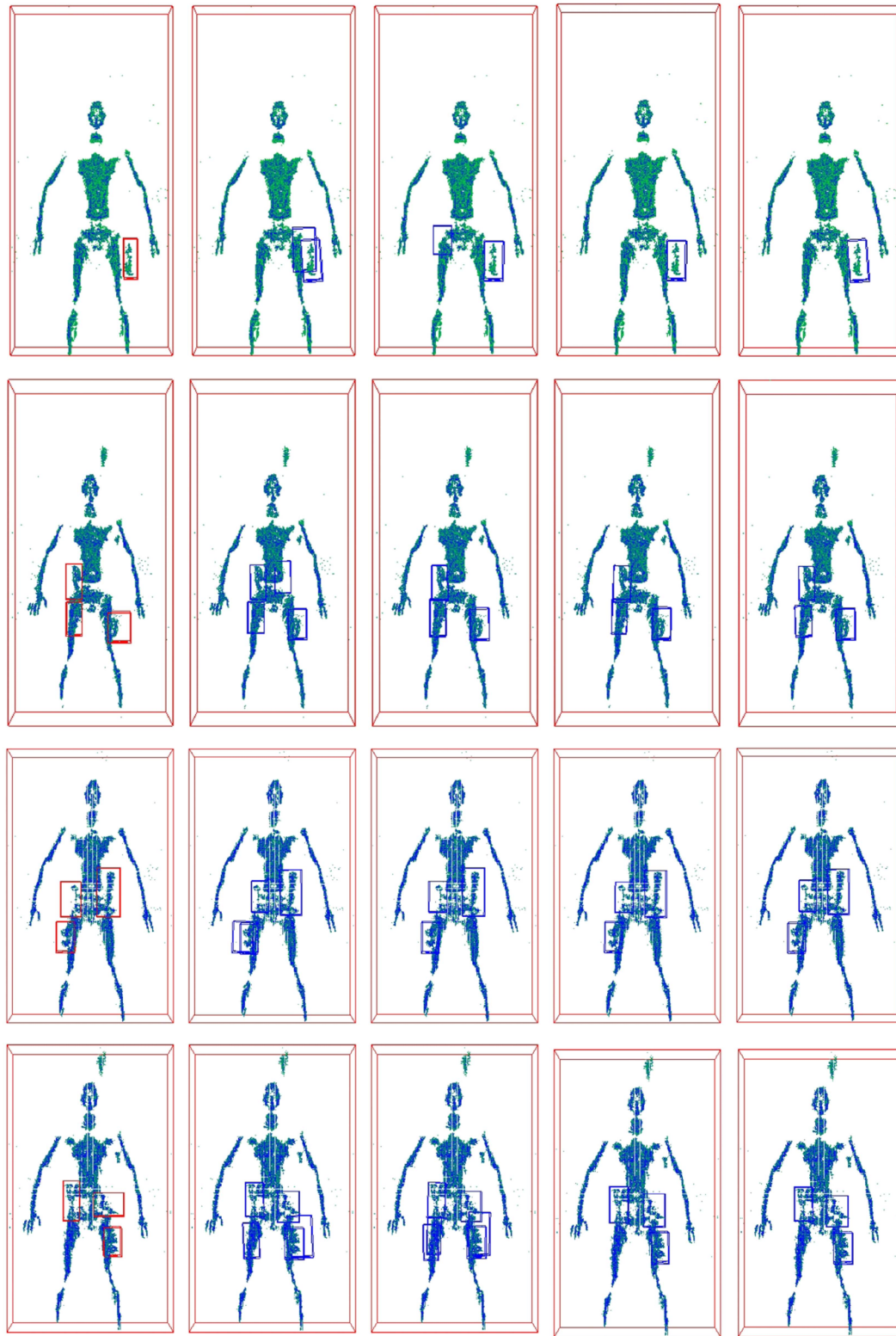


Fig. 8. Detection results. The first column denotes the ground truth. The second column denotes the results by PointPillars. The third column denotes the results by PointPillars with MFE integrated. The fourth column denotes the results by PointPillars with PAFE integrated. The fifth column denotes the results by the proposed method, with MFE and PAFE integrated.

IV. CONCLUSION

This article focuses on adapting the PointPillars network, originally developed for LiDAR point cloud processing, to the realm of 3-D MMW object detection. By

fine-tuning the hyperparameters of this established model to suit MMW imaging contexts, a feature-enhanced PointPillars is proposed, that incorporates feature enhancement techniques within both the pillar feature network and the

backbone network. In the pillar feature network, an MFE module is integrated, which augments the feature aggregation capacity without increasing the parameter count. This allows for a more nuanced representation of point cloud data within each pillar. Additionally, in the backbone network stage, we employ a strategy of PAFE. This approach effectively processes contextual information across varying scales, providing a broadened field of view that facilitates more accurate object detection and localization, particularly for small and medium-sized objects. The experimental evaluations indicate a noteworthy improvement over the baseline model. The enhanced PointPillars algorithm consistently yields high BEV and AOS average precisions of 0.96 and 0.979, respectively. Moreover, the 3-D AP at an IoU threshold of 0.5 is enhanced by 7.16%. With an inference speed of 0.0132 s per frame, the modified network satisfies real-time operational criteria. These findings substantiate the effectiveness of the proposed enhancements in MMW object detection tasks, successfully addressing the initial limitations of the PointPillars model, particularly its inference speed, which previously impeded its practical deployment in MMW detection scenarios.

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